hdoc day 37 by trishil chauhan

Aim

Write a menu-driven C program using separate functions to convert a lowercase string to uppercase without built-in conversion functions, reverse a string, check whether a string is a palindrome, and exit.

C Program

```
#include <stdio.h>

#define MAX 200

void toUppercase(char str[]) {
    int i;
    for (i = 0; str[i] != '\0'; i++) {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - ('a' - 'A');
    }
    printf("Uppercase string: %s\n", str);
}

void reverseString(const char str[]) {
    int len = 0, i;
    while (str[len] != '\0')
        len++;
    printf("Reversed string: ");
    for (i = len - 1; i >= 0; i--)
        putchar(str[i]);
    putchar('\n');
}

void checkPalindrome(const char str[]) {
    int len = 0, left, right, palindrome = 1;
    while (str[len] != '\0')
        len++;
    left = 0;
    right = len - 1;
    while (left < right) {
        if (str[left] != str[right]) {
            palindrome = 0;
            break;
        }
        left++;
        right--;
    }
    if (palindrome)
        printf("The string is a palindrome.\n");
    else
        printf("The string is not a palindrome.\n");
}

int main(void) {
    char str[MAX];
    int choice;
    do {
        printf("\n--- STRING OPERATIONS MENU ---\n");
        printf("1. Convert lowercase to uppercase\n");
        printf("2. Reverse a string\n");
        printf("3. Check palindrome\n");
        printf("4. Exit\n");
```

```
        printf("Enter your choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            return 1;
        }
        if (choice >= 1 && choice <= 3) {
            printf("Enter a string: ");
            scanf(" %199[^\n]", str);
        }
        switch (choice) {
            case 1: toUppercase(str); break;
            case 2: reverseString(str); break;
            case 3: checkPalindrome(str); break;
            case 4: printf("Exiting program.\n"); break;
            default: printf("Invalid choice.\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Output

```
--- STRING OPERATIONS MENU ---
1. Convert lowercase to uppercase
2. Reverse a string
3. Check palindrome
4. Exit
Enter your choice: 1
Enter a string: hello world
Uppercase string: HELLO WORLD

Enter your choice: 2
Enter a string: computer
Reversed string: retupmoc

Enter your choice: 3
Enter a string: madam
The string is a palindrome.

Enter your choice: 4
Exiting program.
```

Result

The menu-driven program successfully performs uppercase conversion, string reversal, and case-sensitive palindrome checking using separate user-defined functions.